PROJECT SUTRA — GROUNDED GRAND FINALS PITCH

REALISTIC FIELD DISASTER DELIVERY • HARIKA (OPERATIONS) • NIKHIL (TECH LEAD)

PRESENTERS: HARIKA • NIKHIL
TOTAL TIME: ~2 MIN 15 SEC

💡 GROUNDED DELIVERY: Connect directly to real Indian disaster audits (Wayanad, Chamoli, Sikkim). Speak the physical reality in plain English while the slides provide the mathematical and algorithmic proof.

SLIDE 01 Introduction: Project SUTRA

00:00 - 00:30 (30s)

SPEAKER: HARIKA (OPERATIONS LEAD) [STAGE: CENTER STAGE • COMPELLING & GROUNDED]

> 🔥 **HOOK: "When landslides hit Wayanad or flash floods tear through Sikkim, rescue teams have a golden 72-hour window to find survivors. But the second conventional drones enter thick forests or mountain gorges, they lose signal and crash."**

"Respected jury, we are **Team OFFGRID**. I am Harika, and with me is Nikhil, our Tech Lead. We built **Project SUTRA** — an autonomous drone swarm engineered to find survivors in disaster zones where **GPS fails and communications collapse**."

"Instead of single manual drones, SUTRA deploys an autonomous swarm backed by 3 breakthroughs:"

- PILLAR 1 **Decentralized Flight:** Drones dodge obstacles and coordinate together at 50Hz without ground towers. `[Slide: 50Hz GNC]`
- PILLAR 2 **Machine-Learning Comms:** Live video stays connected through deep mountain ravines without blacking out. `[Slide: -5dB JSCC]`
- PILLAR 3 **Tri-Modal AI & 3D Vision:** Combines daylight, thermal heat, and radar to pinpoint survivors within 3.5 cm. `[Slide: 3.59cm DEM]`

CONTINUES WITH HARIKA → "Let me show you why today's search and rescue missions fail in the field."

SLIDE 02 The Problem: 4 Critical Failure Voids

00:30 - 01:15 (45s)

SPEAKER: HARIKA [STAGE: URGENT, EMPIRICALLY GROUNDED DISASTER AUDIT]

> ⚠️ **HOOK: "When our team audited actual NDRF after-action reports across Indian disasters, we identified 4 fatal bottlenecks:"**

- VOID 1 (GNC) **Forest Canopies Block GPS:** Dense foliage scatters satellite signals. In the 2024 Wayanad landslides, commercial flight controllers drifted into trees, causing a **70% airframe loss rate**.
- VOID 2 (COMMS) **Mountain Ridges Cut Radio:** In deep valleys like Chamoli, ridgelines sever line-of-sight. Standard digital video drops off a cliff, cutting feeds to total black.
- VOID 3 (VISION) **Flat-Earth Map Errors on Slopes:** Standard drone cameras assume flat ground. On steep 45-degree Himalayan hillsides, coordinates miss by **15 to 30 meters**, routing rescue squads to empty cliffs.
- VOID 4 (C2 GCS) **Human Pilot Bottleneck:** Flying 5 drones requires **15 to 25 ground crew** and takes 2 hours to set up — burning ₹12.5 Lakhs per mission while survivors run out of air.

CONTINUES WITH HARIKA → "Here is how SUTRA's 4 core moats solve each breakdown."

SLIDE 03 The Solution: 4 Moats & 3-4x Performance Delta

01:15 - 02:15 (60s)

SPEAKER: HARIKA [STAGE: HIGH-ENERGY SOLUTION PRESENTATION]

🛡️ **HOOK: "SUTRA solves each void through 4 hardened engineering moats:"**

- MOAT 1 **Self-Flying Swarms:** Drones calculate collision-free flight paths onboard at 50Hz — zero ground towers, zero GPS reliance. [Slide: SUTRA-FSD & 3D ORCA]
- MOAT 2 **ML-Enabled Video Link:** When signals weaken behind mountain ridges, our neural encoder transmits clear video down to -5dB SNR without digital blackouts. [Slide: Deep JSCC | -5dB]
- MOAT 3 **Tri-Modal AI & 3D Slope Mapping:** Fuses daylight vision, body heat thermal sensors, and radar motion to pinpoint survivors within **3.59 cm** in under 15ms. [Slide: 3D DEM Jetson Orin]
- MOAT 4 **One-Person Tactical Command:** An offline 3D tactical map lets **1 operator command the entire swarm** for just **₹95,000 per sortie**. [Slide: Pegasus 3D GCS]

"The bottom line: SUTRA scans a full square mile in 10 to 18 minutes — 3 to 4 times faster than traditional teams, saving lives when every second counts."

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi²	2-3 hours	45-60 min	10-18 min (4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000

🔥 POWER HANDOFF TO NIKHIL → "Now, our Tech Lead, Nikhil, will walk you through the deep engineering behind each subsystem and our real-world field deployment."

STAGE TAKEOVER: NIKHIL (TECH LEAD) [DEEP SUBSYSTEM DEEP DIVES, CODE ARCHITECTURE & FIELD CONOPS]

- **Slide 04 (Subsystem A):** 50Hz PX4 MicroXRCE Offboard + 3D ORCA Velocity Obstacles
- **Slide 05 (Subsystem B):** Deep JSCC Autoencoders & Mesh Resiliency at -5dB SNR
- **Slide 06 (Subsystem C):** Tri-Modal Cross-Attention (RGB + FLIR Thermal + Radar) + WGS84 3D DEM Geolocation
- **Slide 07 (Subsystem D):** WebGPU 3D Digital Twin & Offline ATAK Field Command
- **Slide 08 (Field CONOPS & Q&A):** Wayanad Simulation Audit, Hardware BOM, and Jury Defense